

Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Introduction

Measures of central tendency and measures of dispersion are basic statistical tools used in Data Science to summarize and analyze numerical data. They help understand the average value, spread of the data, and identify unusual observations (outliers).

Given Data

The annual salaries (in ₹ lakhs) of 9 employees are:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median and Mode

The mean is the arithmetic average of all observations.

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Mean = Sum of observations / Number of observations

Sum of salaries

$$4+5+5+6+7+8+8+9+30=82$$

Number of employees

$$n=9$$

Mean

$$= (4+5+5+6+7+8+8+9+30)/9$$

$$= 82/9$$

$$= 9.11$$

Median

Definition

The median is the middle value when the data is arranged in ascending order.

Ordered Data

4, 5, 5, 6, 7, 8, 8, 9, 30

Since there are 9 observations, the middle (5th) value is:

Median = **7 lakhs**

Mode

Definition

The mode is the value that occurs most frequently.

Calculation

5 appears 2 times

8 appears 2 times

Both occur most frequently.

Mode = 5 lakhs and 8 lakhs (Bimodal)

2. Range

Range is the difference between the highest and lowest values.

Maximum salary

$$=30$$

Minimum salary

$$=4$$

Range = Maximum Value – Minimum Value

$$= 30 - 4$$

$$= 26$$

Range = 26

3. Variance and Standard Deviation

Variance measures how far the values are spread from the mean.

$$\sum(x-\bar{x})^2 / 9$$

$$= 56.54$$

Variance = **56.54**

Standard deviation is the square root of variance.

$$\sqrt{56.54} = 7.52$$

Standard Deviation = **7.52**

4. Outlier

Using the IQR method:

$$Q1 = 5$$

$$Q3 = 8.5$$

$$IQR = 3.5$$

$$\text{Upper Limit} = 13.75$$

Since **30 > 13.75**, **30 is the outlier.**

5. Best Measure of Central Tendency

Median (7) is the best measure because the outlier (30) increases the mean to 9.11, making it less representative of the typical salary.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be **right-skewed or left-skewed**.
5. Explain why the median may be more useful than the mean.

Introduction

Website response time is the amount of time a website takes to respond to a user's request. In Data Science, statistical measures such as mean, median, range, variance, and standard deviation help analyze the performance and consistency of response times. These measures also help identify skewness and determine whether extreme values affect the data.

Given Data

The response times (in milliseconds) of a website for 10 requests are:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Mean and Median

Mean is the average response time.

Sum of response times

$$120+125+128+130+132+135+140+145+150+300=1505$$

Number of observations

$$n=10$$

Mean

$$=1505 / 10$$

$$=150.5$$

$$\frac{1505}{10} = 150.5 \text{ ms}$$

Mean = 150.5 ms

Median

Definition

The **median** is the middle value of the ordered data.

Since there are **10 observations (even number)**,

Median

$$=(132+135) / 2$$

$$=267 / 2$$

$$= 133.5$$

Median = 133.5 ms

2. Range

Range is the difference between the highest and lowest values.

Range = Maximum Value – Minimum Value

$$=300-120$$

$$=180$$

Range = 180 ms

3. Variance and Standard Deviation

Variance measures how far the values are spread from the mean.

Variance = **2636.25**

Standard deviation is the square root of variance.

Standard Deviation = **51.34 ms**

4. Skewness

Definition

A distribution is:

- **Right-skewed (positively skewed)** when a few very large values pull the distribution toward the right.
- **Left-skewed (negatively skewed)** when a few very small values pull the distribution toward the left.

Since the value **300** is much larger than the others, the data is **right-skewed (positively skewed)**.

5. Why Median is Better

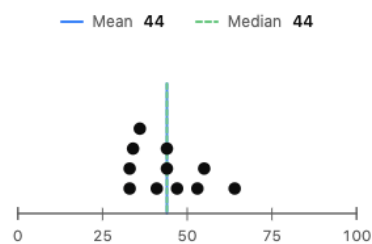
The median is not affected by the unusually high response time (300 ms), so it better represents the typical website response time.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the **IQR**.
3. Use the **$1.5 \times \text{IQR}$ rule** to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

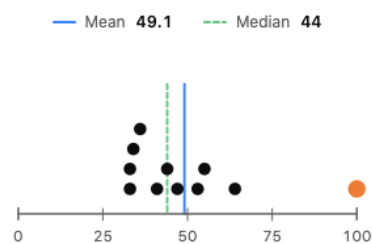


No outlier: mean is 44; median is 44

No outlier

High outlier

New data set



Outlier: mean moves 44 → 49.1; median stays 44

No outlier

High outlier

New data set

Introduction

Student performance can be analyzed using descriptive statistics such as quartiles, interquartile range (IQR), mean, and median. Quartiles divide the data into four equal parts, while the IQR helps identify outliers. Comparing the mean and median shows how extreme values affect the dataset.

Given Data

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2, and Q3

- **Q1 (First Quartile):** The median of the lower half of the data.
- **Q2 (Second Quartile):** The median of the entire dataset.
- **Q3 (Third Quartile):** The median of the upper half of the data.

Ordered Data

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Q2 (Median)

Since there are **12 observations (even number)**:

$$Q2 = (52+55) / 2 = 53.5$$

Q1

Lower half:

42, 45, 48, 50, 52, 52

$$Q1 = (48+50) / 2 = 49$$

Q3

Upper half:

55, 58, 60, 62, 65, 95

$$Q3 = (60+62) / 2 = 61$$

2. Calculate the IQR

Definition

The Interquartile Range (IQR) measures the spread of the middle 50% of the data.

Formula

$$\text{IQR} = Q3 - Q1$$

Calculation

$$\text{IQR} = 61 - 49 = 12$$

Answer

$$\text{IQR} = 12$$

3. Use the $1.5 \times \text{IQR}$ Rule to Identify Any Outliers

An **outlier** is a value that lies far from the rest of the observations.

Formula

Lower Limit

$$Q1 - 1.5(\text{IQR})$$

Upper Limit

$$Q3 + 1.5(\text{IQR})$$

Calculation

Lower Limit

$$49 - (1.5 \times 12) = 49 - 18 = 31$$

Upper Limit

$$61 + (1.5 \times 12) = 61 + 18 = 79$$

Any value less than **31** or greater than **79** is an outlier.

Since, $95 > 79$

95 is an outlier.

4. Find the Mean and Median

Mean

The **mean** is the average of all observations.

Formula

$$\text{Mean} = \sum x / n$$

Calculation

Sum of marks

$$42+45+48+50+52+52+55+58+60+62+65+95=684$$

Number of students

$$n = 12$$

Mean

$$= 684 / 12 = 57$$

$$\text{Mean} = 57$$

Median

Median is the middle value.

Median

$$= (52+55) / 2$$

$$= 53.5$$

$$\text{Median} = 53.5$$

5. Effect of the Outlier

The outlier **95** increases the mean from the typical marks, while the median changes very little. Therefore, the median better represents the students' performance.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Introduction

In an e-commerce business, analyzing daily order data helps understand customer demand and business performance. Statistical measures such as mean, median, mode, variance, standard deviation, and coefficient of variation are useful for measuring the average number of orders and the variability in daily sales.

Given Data

The number of orders received over 10 days is:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the Mean, Median, and Mode

Mean

The **mean** is the arithmetic average of all observations.

Formula

$$\text{Mean} = \sum x/n$$

Sum of orders

$$18+20+22+22+24+25+26+28+30+35=250$$

Number of observations

$$n=10$$

Mean

$$= 250 / 10 = 25$$

Mean = 25 orders

Median

The **median** is the middle value of the ordered data.

Since there are **10 observations**, the median is the average of the 5th and 6th values.

$$\text{Median} = (24 + 25) / 2 = 24.5$$

Median = 24.5 orders

Mode

The **mode** is the value that appears most frequently.

From the data,

22 appears **twice**, while all other values appear once.

Mode = 22 orders

2. Find the Variance and Standard Deviation

Variance

Variance measures how much the observations are spread around the mean.

Formula

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

Using the given data,

Variance = 19.40

Standard Deviation

Standard deviation is the square root of the variance.

Formula

$$SD = \sqrt{\text{Variance}}$$
$$= \sqrt{19.40} = 4.40$$

Variance = **19.40**

Standard Deviation = **4.40 orders**

3. Calculate the Coefficient of Variation

The Coefficient of Variation (CV) measures the relative variability of the data.

Formula

$$CV = \text{Standard Deviation} / \text{Mean} \times 100$$

Calculation

$$CV = 4.40 / 25 \times 100 = 17.6\%$$

Coefficient of Variation = 17.6%

4. What Does the Standard Deviation Tell the Company?

The standard deviation measures how much the daily orders vary from the average.

A standard deviation of **4.40** means that the daily number of orders usually differs from the average by about **4 to 5 orders**. This helps the company understand how consistent customer demand is and supports inventory and staffing decisions.

5. Would You Describe the Order Data as Highly or Moderately Variable?

The coefficient of variation is **17.6%**, which indicates **moderate variability**.

The daily order values are fairly close to the average, showing that customer demand is relatively stable and predictable.

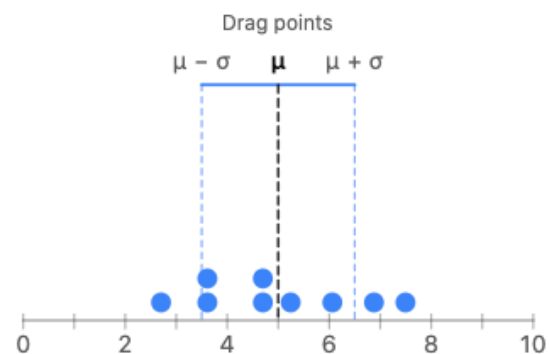
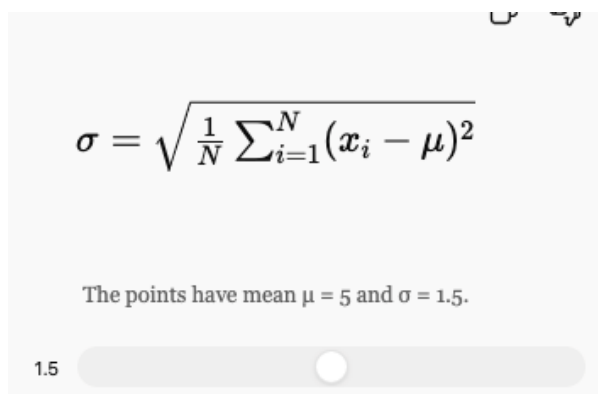
Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.



Introduction

In Machine Learning, prediction errors measure how far the predicted values are from the actual values. Statistical measures such as mean, variance, and standard deviation help compare the performance and consistency of different models. A model with lower variability in errors is generally considered more reliable.

Given Data

Model A Prediction Errors

2, 3, 3, 4, 4, 5, 5, 6

Model B Prediction Errors

1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the Mean Error for Both Models

The **mean error** is the average prediction error made by a model.

Formula

$$\text{Mean} = \sum x / n$$

Model A

Sum of errors

$$2+3+3+4+4+5+5+6=32$$

Number of observations

$$n=8$$

Mean

$$=32 / 8 = 4$$

$$\text{Mean Error (Model A)} = 4$$

Model B

Sum of errors

$$1+1+2+4+5+7+8+9=37$$

Mean

$$=37/8=4.625$$

$$\text{Mean Error (Model B)} = 4.625$$

2. Calculate the Variance for Both Models

Definition

Variance measures how much the prediction errors are spread around the mean.

Formula

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

Calculation

Model A

Variance = **1.50**

Model B

Variance = **9.23**

3. Calculate the Standard Deviation for Both Models

Definition

Standard deviation is the square root of the variance and indicates how much the errors vary from the average error.

Formula

$$SD = \sqrt{\text{Variance}}$$

Calculation

Model A

$$SD = \sqrt{1.50} = 1.22$$

Model B

$$SD = \sqrt{9.23} = 3.04$$

4. Which Model Has More Consistent Predictions?

A model with a **lower standard deviation** has more consistent prediction errors.

Since

- Model A Standard Deviation = **1.22**
- Model B Standard Deviation = **3.04**

Model A is more consistent because its prediction errors are clustered closer to the mean.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain.

No.

A lower mean error alone does not guarantee that a model is better.

A good machine learning model should have:

- Low average prediction error.
- Low variance and standard deviation.
- Consistent predictions on different datasets.
- Good generalization to new, unseen data.

Therefore, while the mean error is important, consistency and reliability should also be considered when evaluating model performance.